Plan B: E-MS

A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF MASTER OF PUBLIC HEALTH DATASCIENCE

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Abstract

Missing data remains a persistent challenge across fields such as genetics, epidemiology, and social sciences. One critical yet under-addressed issue in the treatment of incomplete data is double dipping, the inappropriate reuse of data in both model selection and inference (Jiang, Nguyen, and Rao 2015, 2011). This work investigates how double dipping manifests in the context of missing data imputation and demonstrates that the Expectation-Maximization-Model Selection (E-MS) framework (Jiang, Nguyen, and Rao 2015) effectively addressing aspects of this problem. We compare E-MS to conventional methods across a range of data-generating scenarios under different missing data mechanisms, including missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR). Large-scale simulations are conducted to assess the robustness of E-MS and to evaluate its performance relative to existing approaches. Results demonstrate that E-MS mitigates key biases associated with double dipping, consistently outperforming conventional methods by achieving superior F1 scores through high recall coupled with strong precision.

1 Introduction

In applied statistical analysis, missing data is an unavoidable and often complex obstacle. Whether due to non-response in surveys, sensor failure in experiments, or limitations in sequencing technologies in genomics, analysts must frequently address datasets with incomplete observations. The implications of improper handling of missingness are far-reaching, especially when decisions about model structure or inference procedures are based on incomplete information.

This paper focuses on a particular consequence of missing data analysis double dipping (Jiang, Nguyen, and Rao 2015, 2011). This occurs when the same data is used in multiple stages of model development without proper adjustment, typically in both model selection and inference. Such practices can inflate Type I error rates and lead to misleading conclusions.

The goal of this work is to understand how imputation methods, particularly the E-MS algorithm, interact with and potentially exacerbate or alleviate double dipping.

Prior work has shown that standard information criteria applied directly to incomplete logistic regression data can systematically favor underspecified models. For example, (Ibrahim, Zhu, and Tang 2008) demonstrate that, under a missing at random mechanism, the intercept-only model may maximize the conditional expected log-likelihood even when it is not the true model. This phenomenon can lead traditional BIC-based selection procedures, such as the BIC penalty denoted by IZT, to consistently choose an incorrect model. Through simulations across different sample sizes, true model structures, and both ignorable and nonignorable missingness mechanisms, they show that alternative methods like E-MS achieve substantially higher true positive rates and are more robust to the missingness mechanism (Ibrahim, Zhu, and Tang 2008; Rubin 1976).

2 Missing Data

To understand the challenges of missing data, we must first distinguish among the common mechanisms by which data may be missing. Missing at Random (MAR) and Missing not at Random (MNAR) (Rubin 1976) are concepts crucial in understanding the nature of missing data in statistical analysis. MAR refers to a situation where the probability of missing data depends only on observed data and not on unobserved (missing) data. In other words, once we account for the observed variables, any remaining missing data is random. T his allows for unbiased estimation if the missingness is properly accounted for in the analysis. On the other hand, MNAR occurs when the probability of missing data depends on the unobserved data itself, even after accounting for the observed data. This introduces potential bias because the missingness is related to the values of the missing data, making it more challenging to handle in statistical analyses without making strong assumptions about the missing data mechanism. Understanding whether data is MAR or MNAR is crucial for

applying appropriate statistical methods and drawing valid conclusions from incomplete datasets.

While, Missing Completely at Random (MCAR), would be a special case where missingness is entirely independent of both observed and unobserved data, an assumption rarely met in real-world data.

Incomplete cases, particularly those with only 20-40% of the full information available, often appear in genetic studies such as Quantitative Trait Loci (QTL) mapping, where limitations in data collection or sequencing coverage lead to substantial gaps. While several approaches exist to handle such missing data, from complete case analysis to multiple imputation, few adequately address the implications of using imputed data during both model selection and final inference.

3 Imputation

Single imputation methods replace each missing value with one deterministic estimate. Although these approaches are easy to implement and computationally efficient, they often fail to capture the uncertainty inherent in missing data.One of the most widely used approaches is mean, median, or mode imputation, in which a missing value is replaced with a measure of central tendency derived from the observed data. While this method preserves sample size and is straightforward to apply, it artificially reduces variability and can bias parameter estimates, especially when the data are not missing completely at random (Little and Rubin 2002).

Another form of single imputation is regression imputation, where missing values are predicted using a regression model built from the observed cases. Although regression-based approaches provide more plausible values than simple mean imputation, they still tend to overstate the precision of the data by treating predicted values as known quantities (Allison 2011).

Multiple imputation, introduced by (Rubin 1987), is widely considered the gold standard for handling missing data in many applied research contexts. Instead of substituting missing values with a single estimate, MI generates multiple versions of the dataset, each containing different plausible imputations drawn from a predictive distribution that reflects both observed data and uncertainty.

After statistical analyses are performed separately on each imputed dataset, the results are combined using Rubin's rules, which account for both within-imputation and between-imputation variability. This leads to unbiased parameter estimates and valid standard errors when data are missing at random (MAR) Buuren (2018). Although MI requires more computation and careful model specification than single imputation, its ability to incorporate uncertainty makes it far more statistically robust.

Model-based imputation methods leverage statistical or machine learning models to estimate missing values in a data-driven manner. These approaches are often more flexible than traditional single or multiple imputation.

A foundational model-based method is the Expectation–Maximization (EM) algorithm, which iteratively estimates missing data by maximizing the likelihood of the observed data under assumed model parameters (Dempster, Laird, and Rubin 1977). EM produces maximum likelihood estimates but does not generate multiple datasets, so it does not account for imputation uncertainty unless paired with MI.

Machine-learning–oriented methods have become increasingly popular. k-Nearest Neighbors (kNN) imputation identifies cases with similar feature profiles and uses their observed values to fill in missing entries. Random forest–based methods, such as missForest, build ensembles of decision trees to predict missing values; they can handle nonlinear relationships and mixed data types effectively (Stekhoven and Bühlmann 2012). More recently, deep learning autoencoder imputation has been used to reconstruct missing values by learning compressed representations of the full dataset, making it particularly effective in high-dimensional or complex data structures.

Model-based imputation techniques offer strong predictive performance and flexibility, especially in modern large-scale datasets. However, they require careful tuning and may not inherently provide valid measures of uncertainty unless extended within a multiple-imputation framework.

4 Double Dipping problem

In high-dimensional QTL studies and related genetic mapping contexts, a common analytical workflow uses the same dataset both to select models and to estimate their parameters or significance. This practice, often referred to as double dipping or circular analysis, arises when data are used first to generate hypotheses or select features and then reused to estimate or test those features without accounting for the selection process. The statistical danger of this workflow has been well documented in other fields, particularly in neuroimaging, where it was shown that re-analyzing data after selection leads to systematically biased effect sizes and invalid inferential statements (Kriegeskorte et al. (2009)). QTL mapping is a particularly acute instance of this problem. Because the number of candidate loci vastly exceeds the number of observations, researchers must scan across the genome and select regions that exceed some significance threshold — yet most classical statistical inference assumes that the model was specified independently of the data used to estimate it. In QTL mapping, this assumption is routinely violated and the impact is magnified by the scale, sparsity, and missingness that characterize such data.

Markers, covariates, and interaction structures are typically chosen using data-dependent criteria such as likelihood scans, penalized regression, or information criteria, and the same data are then used to estimate effect sizes and uncertainties for the selected loci. Conditioning on the outcome in this way induces selection bias: the chosen loci are those that happened to align with random fluctuations, imputation artifacts, or sampling noise, and treating the selected model as fixed ignores this conditioning. As a result, estimated genetic effects are

biased away from zero and uncertainty is understated, even when model selection is performed using principled criteria rather than hypothesis testing (Pasaniuc and Price (2017)).

This issue becomes more severe in the presence of missing data and genotype imputation, which are now unavoidable in modern QTL studies. High-resolution mapping pipelines rely on probabilistic reconstruction of unobserved genotypes from low-coverage sequencing or from related individuals, often in complex multi-parent populations (Huang and al. (2012); Karl W. Broman et al. (2017)). These imputation models are trained on the observed data and encode assumptions about haplotype structure, recombination, and linkage disequilibrium. When the imputed values are then treated as fixed inputs to downstream QTL scans, the uncertainty and model dependence introduced during imputation are effectively ignored. If marker selection or model fitting is carried out on these imputed datasets, the selection step itself is influenced by the imputation model, which further entangles model choice with the observed data and amplifies double-dipping bias. In this sense, sophisticated imputation does not remove the inferential problem; it relocates it into a more complex, and often hidden, modeling layer.

The practical response to multiple testing in QTL studies is rarely to rely on strict p-value thresholds alone, because the number of candidate loci is enormous and effect sizes are typically small. Instead, researchers often adopt penalized likelihoods, Bayesian model probabilities, or criteria such as BIC to balance goodness-of-fit against model complexity. While these tools are well suited for high-dimensional search, they do not by themselves resolve the inferential consequences of data-dependent model choice. Once a model is selected, standard errors, posterior summaries, or test statistics are usually computed as if the model were fixed in advance, even though it was chosen precisely because it fit the data unusually well. This creates a false sense of certainty about the importance of specific loci, and it is one of the main drivers of the winner's-curse and reproducibility problems in genetic association studies.

Recent methodological surveys in high-dimensional biomedical inference emphasize that

many modern pipelines successfully address feature selection, prediction, and missingness, yet still leave unresolved the deeper statistical issue of how to perform valid inference after selection (Serghiou et al. (2023)). In QTL mapping, where model search, imputation, and inference are tightly interwoven, this gap is particularly consequential. Double dipping is therefore not a technical detail about p-values versus information criteria; it is a structural problem that arises whenever the same data are used to decide what model to believe and to quantify how strongly that model is supported. Without accounting for this dependency, QTL analyses can produce inflated effect estimates, underestimated uncertainty, and spurious discoveries, even when every individual modeling step appears statistically sound.

5 Existing Approaches

5.1 LASSO

The Least Absolute Shrinkage and Selection Operator (Lasso), introduced by (Tibshirani 1996), is a regularization-based regression method designed for settings where model interpretability and prediction accuracy are both important, particularly in high-dimensional regimes where the number of predictors p may be comparable to or larger than the sample size n. Lasso extends ordinary least squares by adding an ℓ_1 penalty on the regression coefficients, which encourages sparsity and enables simultaneous variable selection and estimation.

Formally, suppose the data follow the linear regression model

$$y = X\beta + \varepsilon,$$

where $y \in \mathbb{R}^n$ denotes the response vector, $X \in \mathbb{R}^{n \times p}$ is the design matrix of predictors, $\beta \in \mathbb{R}^p$ is the vector of unknown regression coefficients, and ε represents random error. The

Lasso estimator is obtained by solving the penalized least-squares problem

$$\hat{\beta}^{\text{lasso}} = \arg\min_{\beta} \left\{ \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 \right\}$$

where $|\beta|1 = \sum j = 1^p |\beta_j|$ denotes the ℓ_1 norm and $\lambda \geq 0$ is a tuning parameter that controls the degree of regularization. Larger values of λ impose stronger shrinkage, forcing more coefficients to be exactly zero and producing a sparse model that retains only the most relevant predictors.

This sparsity-inducing property makes Lasso particularly appealing for imputation and prediction tasks in complex data environments. When imputing missing values, Lasso can identify a small set of informative predictors that contribute most strongly to the conditional expectation of the missing variable, thereby reducing variance and mitigating overfitting. More broadly, Lasso is widely used in genomics, finance, and machine learning for handling multicollinearity, improving prediction accuracy, and producing interpretable models in high-dimensional settings.

Despite these advantages, standard Lasso formulations struggle in the presence of missing data. Classical Lasso assumes that both the response vector y and the design matrix X are fully observed. In practice, widely used implementations require complete data matrices, and will either fail or implicitly discard observations containing missing values. Unlike certain tree-based or ensemble methods, Lasso does not natively incorporate mechanisms for handling missingness during model fitting.

As a consequence, the quality of any preprocessing step becomes critical. If missing values are imputed poorly before applying Lasso, the resulting coefficient estimates and variable selection can be severely biased. Since the ℓ_1 penalty is sensitive to the scale and structure of the predictors, errors introduced during imputation may lead to incorrect exclusion of relevant variables or inclusion of spurious ones.

Several workarounds are commonly employed in practice. The most straightforward

approach is to apply an imputation procedure prior to fitting the Lasso model, using methods such as mean imputation, k-nearest neighbors, or more sophisticated approaches like multiple imputation by chained equations (Buuren and Groothuis-Oudshoorn 2011). Alternatively, research has proposed modified versions of Lasso that explicitly account for missing data, including EM-based Lasso formulations or joint models that combine imputation and coefficient estimation within a unified framework. While these extensions offer theoretical improvements, they are not yet standard in most applied software.

5.2 QTL

Quantitative trait locus (QTL) mapping seeks to identify genomic regions that are associated with variation in a quantitative phenotype. In practice, QTL analysis must account for missing or unobserved genotype information between genetic markers, which arises naturally due to limited marker density. The `R/qtl` framework, as developed and documented by (Karl W. Broman et al. 2019) and the `rqtl` manual, provides a comprehensive set of tools for addressing this problem using likelihood- and regression-based approaches.

To identify markers of interest, QTL mapping is commonly performed using three complementary methods: expectation–maximization (EM) algorithms (Dempster, Laird, and Rubin 1977; Lander and Botstein 1989), Haley–Knott (HK) regression (Haley and Knott 1992), and multiple imputation approaches (Sugiyama et al. 2001). All three methods rely on the same underlying genetic model but differ in how they handle uncertainty in the unobserved genotypes at putative QTL positions.

The EM-based approach treats the underlying QTL genotypes as latent variables. At each genomic position, the E-step computes the conditional probabilities of the possible QTL genotypes given the observed marker data and current parameter estimates, while the M-step maximizes the expected complete-data log-likelihood to update model parameters. This method provides statistically efficient inference but can be computationally intensive, particularly for large genomes or complex crosses.

Haley–Knott regression offers a computationally simpler approximation to the EM algorithm. Instead of fully iterating between expectation and maximization, HK regression replaces the unobserved QTL genotypes with their conditional expectations and fits a standard linear regression model. While this approximation leads to substantial gains in speed, it may result in a modest loss of efficiency compared to the full EM approach, especially in regions with sparse marker information.

Multiple imputation methods take a different perspective by repeatedly sampling complete genotype datasets from the conditional distribution of the missing genotypes given the observed markers. For each imputed dataset, a standard QTL scan is performed, and the results are combined across imputations. This strategy more fully propagates uncertainty in the missing genotypes and often yields more robust inference, at the cost of increased computational effort.

In genomic applications, QTL mapping is typically conducted on large-scale datasets involving thousands of genetic markers and, in modern studies, potentially thousands of individuals. In such settings, even small systematic errors in the handling or imputation of missing genotype data can accumulate and substantially bias downstream inference. Poorly specified imputation models may distort conditional genotype probabilities, inflate or attenuate LOD scores, and lead to spurious QTL detection or missed true signals. Consequently, careful treatment of missing data is critical, as incorrect imputation can greatly skew results despite large sample sizes that might otherwise suggest statistical robustness.

For each mapping method, a genome-wide scan yields a profile of logarithm of the odds (LOD) scores that measure the strength of evidence for the presence of a QTL at each genomic position. Statistical significance is assessed by applying a predefined LOD threshold to these scan results, and all genomic locations where the LOD score exceeds this threshold are identified as candidate QTL. Reporting all detected peaks across the genome for each method enables a systematic comparison of the number, location, and strength of QTL signals, facilitating evaluation of detection consistency and relative mapping performance

among the EM, Haley–Knott, and multiple imputation approaches.

5.3 EM - The Expectation-Maximization

The Expectation–Maximization (EM) algorithm, introduced by (Dempster, Laird, and Rubin 1977), provides a likelihood-based framework for parameter estimation in the presence of missing or latent data. Its appeal lies in its conceptual simplicity and broad applicability across a wide range of statistical models. Let Y_o denote the observed data, Y_m the missing or latent data, and θ the vector of unknown parameters. If the complete data $Y = (Y_o, Y_m)$ were fully observed, inference would proceed by maximizing the complete-data log-likelihood $\ell_c(\theta) = \log p(Y_o, Y_m \mid \theta)$. Because Y_m is unobserved, direct maximization of the observed-data likelihood $\ell(\theta) = \log p(Y_o \mid \theta)$ is often analytically intractable. EM addresses this difficulty by iteratively maximizing a surrogate objective function that is easier to handle.

At iteration t, the E-step constructs the function

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{Y_m \mid Y_o, \theta^{(t)}} \Big[\log p(Y_o, Y_m \mid \theta) \Big],$$

which is the conditional expectation of the complete-data log-likelihood with respect to the distribution of the missing data given the observed data and the current parameter estimates $\theta^{(t)}$. This step does not impute missing values deterministically; instead, it averages over their conditional distribution under the current model, thereby producing an expected log-likelihood that reflects the uncertainty induced by missingness, but only through expectation rather than full probabilistic propagation.

The M-step then updates the parameter estimates by maximizing this expected log-likelihood,

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(t)}).$$

This optimization yields new parameter values that increase, or at least do not decrease, the observed-data likelihood. Repeating the E-step and M-step generates a sequence $\theta^{(t)}$ with the

property that $\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})$, and the algorithm is guaranteed to converge to a stationary point of the likelihood under mild regularity conditions.

Despite its strengths, EM has several important limitations. First, it assumes that the model specified for the complete data is correct; any model misspecification directly affects both the conditional distribution used in the E-step and the optimization performed in the M-step. Second, although the E-step computes expectations over the missing data, EM ultimately treats these expectations as fixed quantities in the M-step, and therefore does not fully propagate uncertainty in the latent variables into parameter uncertainty. Third, the likelihood surface in models with latent structure is often multimodal, and EM may converge to a local rather than global maximum depending on initialization.

6 E-MS

The E–M is well known for parameter estimation in the presence of missing data. On the other hand, model selection, as another component of model identification, may also be viewed as parameter estimation, with the parameter being the (identification (ID) number of) the model and the parameter space being the (ID numbers of the) model space. Namely, we combine the parameters with the model under which the parameters are defined. So, at the current stage of the iteration, we have the current model, M_c, as well as the current estimates of the parameters, $\hat{\theta}_c$, under M_c. Let $Q(M) = Q(Y, M, \theta_M)$ be a measure of lack-of-fit, where Y represents the complete data, M a candidate model, and θ_M the vector of parameters under M. We take the conditional expectation of $Q(M)$ under M_c, with the parameters under M_c, $\theta_{M_c} \equiv \hat{\theta}_c$, given the observed data y_o, denoted by $\mathbb{E}_c\Big[Q(M) \mid y_o\Big]$. This is the E-step.

In the next step, we carry out model selection using$\mathbb{E}_c\{Q(M) \mid y_o\}$ as the measure of lack-of-fit. To do so, we first find $\hat{Q}_c(M) = \inf_{\theta_M \in \Theta_M} \mathbb{E}_c\{Q(M) \mid y_o\}$, where Θ_M is the parameter space under M. We can use $\hat{Q}_c(M)$ in a GIC setting, in which the optimal model, $\hat{M}_{\text{opt}}$, is found by minimizing $\hat{Q}_c(M) + \lambda_n|M|$ over $M \in \mathcal{M}$, the class of candidate models,

where λ_n is a penalty that depends on the sample size n, and $|M|$ is the dimension of M. Alternatively, we may use the fence method (Jiang et al. 2008) based on $\hat{Q}_c(M)$. This is the MS-step, where MS stands for "model selection." We then replace M_c by $\hat{M}_{\text{opt}}$, found in the MS-step, and $\hat{\theta}_c$ by $\hat{\theta}_{\text{opt}}$, where $\hat{\theta}_{\text{opt}}$ is the parameter vector under $\hat{M}_{\text{opt}}$ corresponding to the minimizer of $\mathbb{E}_c\{Q(\hat{M}_{\text{opt}}) \mid y_o\}$ over $\theta_{M_{\text{opt}}} \in \Theta_{\hat{M}_{\text{opt}}}$, and return to the E-step. We illustrate the EM–MS procedure with some examples.

Following (Karl W. Broman and Speed 2002), we have a conditional linear regression model for the phenotype variable Y_i such that, given the marker indicators x_i, we have $Y_i = \sum_{k=1}^{r} \sum_{j \in M_k} \beta_{kj} X_{ijk} + \varepsilon_i$, where r is the number of chromosomes, M_k is a subset of $\{1, \ldots, q\}$, and q is the number of markers on each chromosome, and ε_i is a normal error with mean zero and unknown variance σ^2. The ε_i's are uncorrelated and also independent with the X_{ijk}'s. Furthermore, the marker indicators X_{ijk} are assumed to be a Markov chain within each chromosome, with $P(X_{i1k} = 0) = P(X_{i1k} = 1) = \frac{1}{2}$ (Mendel's rule), and $P(X_{i,j+1,k} = 1 \mid X_{ijk} = 0) = P(X_{i,j+1,k} = 0 \mid X_{ijk} = 1) = \theta$, where θ is the recombination fraction. The problem of interest is to identify the subset $M = (M_1, \ldots, M_r)$, which is viewed as a model selection problem as in (Karl W. Broman and Speed 2002).

We consider the E–MS in conjunction with the BIC procedure. Due to the high dimensionality, we consider the forward/backward (FW/BW) BIC procedure of (Karl W. Broman and Speed 2002). A detailed description of the latter is given in the online supplemental material (Section A.3). The log-likelihood, under a given model M, can be expressed as $\ell_M = \ell_{M,y|x} + \ell_x$, where ℓ_x does not depend on the model. Here, $\ell_{M,y|x} = -0.5(n \log \sigma^2 + \sigma^{-2} \sum_{i=1}^{n} (y_i - x_i' \beta_M)^2)$, c being a constant. Thus, we have $\text{BIC}(M) = -2\ell_M + |M| \log(n)$, where $\hat{\ell}_M$ is the maximized ℓ_M with respect to M and the parameters. It is easy to show that the MLE of θ, $\hat{\theta}$, is the same as the maximizer of ℓ_x, which does not depend on M. Thus, we have ...

$$\text{BIC}(M) = -2\hat{\ell}_{M,\,y|x} + |M| \log(n) - 2\hat{\ell}_x \propto -2\hat{\ell}_{M,\,y|x} + |M| \log(n). \tag{1}$$

In addition, the FW/BW requires evaluation of $\min_\beta \text{RSS}(y, X, \beta)$, where

$$\mathrm{RSS}(y, X, \beta) = \sum_{i=1}^{n} (y_i - x_i'\beta)^2. \tag{2}$$

with $X = (x_i')_{1 \le i \le n}$. Because both (1) and (2) involve missing data, we replace them by their conditional expectations under the current model M_c and the current parameter estimates under M_c, before the minimization/maximization. This leads to

$$\mathrm{RSS}_c(M \mid y_o, x_o) = \min_{\beta} \mathbb{E}_c[\mathrm{RSS}(Y, X, \beta) \mid y_o, x_o], \tag{3}$$

$$\mathrm{BIC}_c(M \mid y_o, x_o) = -2 \max_{\beta, \sigma^2} \mathbb{E}_c[\ell(M, Y \mid X) \mid y_o, x_o] + |M| \log(n). \tag{4}$$

Both of which have closed-form expressions , where x_o and y_o denote the observed x's and y's, respectively.

In summary, given M_c and the current parameter estimates, the FW/BW, based on (3), is used to generate a sequence of models; the BIC, based on (4), is then applied to the sequence generated by the FW/BW to update the model as well as parameter estimates.

A reasonable initial model is the full model, M_f. A reasonable initial estimator for θ is $\hat{\theta}_0$, the proportion of observed cases in which $x_{i,j+1,k}$ and x_{ijk} are different. As for the initial estimator of $\hat{\beta}_f$, the vector of regression coefficients under M_f, note that the idea of least squares (LS) fit in regression is to find the parameter estimates that minimize $\sum_{i=1}^{n}(y_i - \mathbb{E}_f(Y_i \mid x_o)^2$, where $\mathbb{E}_f$ denotes expectation under M_f. Due to the missing data, it is natural to replace this by $\sum_{i=1}^{n}(y_i - \mathbb{E}_f(Y_i \mid x_i))^2$ by $\sum_{i \in I_o}(y_i - \mathbb{E}_f(Y_i \mid x_{i,o}))^2$, where I_o denotes the subset of indices i such that y_i is observed. Furthermore, we have $\mathbb{E}_f(Y_i \mid x_o) = \sum_{k=1}^{r}\sum_{j=1}^{q} \beta_{jk}\, \mathbb{E}_f(X_{ijk} \mid x_{o,i})$, where $x_{i,o}$ denotes the observed x's for the i-th subject, and an expression of the conditional expectation can be easily obtained, with $\hat{\theta}_0$ replacing θ, if x_{ijk} is missing. We then run the LS with y_i, $i \in I_o$, as the responses and $\mathbb{E}_f(X_{ijk} \mid x_{o,i})$, $j = 1, \ldots, q$, as the predictors, to obtain the initial estimator $\hat{beta}_{f,0}$ for β_f. The initial estimator for $\sigma^2, \hat{\sigma_0^2}$, is the RSS of this LS fit divided by $|I_o| - qr$

Let y, x, x_c, x_d denote the data for $y_i, x_i, x_{i,c}, x_{i,d}$, respectively, across $1 \le i \le n$. Then, it can be shown that the complete-data log-likelihood has the expression

$$\begin{aligned} \ell = c - \frac{n}{2}(\log\sigma^2 + \log|\Omega|) + \sum_{r=1}^{s} n_r \log\pi_r \\ -\frac{1}{2}\sum_{r=1}^{s}\sum_{i=1}^{n} \mathbf{1}_{(x_{i,d}=v_r)}(x_{i,c}-\mu_r)'\Omega^{-1}(x_{i,c}-\mu_r) \\ -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - x'_{i,M}\beta_M)^2, \end{aligned} \tag{5}$$

where c is a constant. Note that the maximum likelihood is a constrained maximization problem, namely, maximize ℓ subject to $\sum_{r=1}^{s}\pi_r = 1$. Define $\mathcal{L} = \ell + \lambda\left(\sum_{r=1}^{s}\pi_r - 1\right)$. Then, the MLE of the parameters, plus the Lagrange multiplier λ, is a stationary point of $\mathcal{L}$.

To see an alternative, we now consider the E–MS in conjunction with the adaptive fence (Jiang et al. 2008). See (Jiang 2014) for a recent review of the fence methods. Take the initial model, M_0, as the full model, M_f, and let β_f be β under M_f. Let $\mathbb{E}_f$ denote the conditional expectation under M_f and the current estimates of parameters under M_f, including β_f, σ^2, μ_r, π_r, $r = 1 \le r \le s$, and Ω. Let y_o, x_o denote the observed y, x, respectively. By (5), with $M = M_f$, we have

$$\begin{aligned} \bar{\ell}_f = \mathbb{E}_f(\ell \mid y_o, x_o) = c - \frac{n}{2}\log\sigma^2 \\ - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\mathbb{E}_f\left[(y_i - X_i'\beta_f)^2 \mid y_o, x_o\right], \end{aligned} \tag{6}$$

where c does not depend on β_f and σ^2. From (6), we obtain the updates for β_f and σ^2,

$$\hat{\beta}_f = S_2^{-1}S_1, \qquad \hat{\sigma}^2 = n^{-1}\Big(S_0 - S_1'S_2^{-1}S_1\Big). \tag{7}$$

The quantities in (7) are

$S_0 = \sum_{i=1}^{n}\mathbb{E}_f(Y_i^2 \mid y_o, x_o)$, $S_1 = \sum_{i=1}^{n}\mathbb{E}_f(X_iY_i \mid y_o, x_o)$, $S_2 = \sum_{i=1}^{n}\mathbb{E}_f(X_iX_i' \mid y_o, x_o)$. Furthermore, we have (see Section A.4.2 of the online supplementary material from (Jiang, Nguyen, and Rao 2015))

$$\begin{aligned}\hat{\mu}_r &= \frac{\sum_{i=1}^{n} \mathbb{E}_f\left[\mathbf{1}_{(X_{i,d}=v_r)}\, X_{i,c} \mid y_o, x_o\right]}{\sum_{i=1}^{n} P_f(X_{i,d}=v_r \mid y_o, x_o)},\\ \hat{\pi}_r &= \frac{\mathbb{E}_f(N_r \mid y_o, x_o)}{\sum_{r=1}^{s} \mathbb{E}_f(N_r \mid y_o, x_o)}, \qquad 1 \le r \le s.\end{aligned} \tag{8}$$

$$\hat{\Omega} = \frac{1}{n}\sum_{r=1}^{s}\sum_{i=1}^{n} \mathbb{E}_f\left[\mathbf{1}_{(X_{i,d}=v_r)}\,(X_{i,c}-\hat{\mu}_r)\,(X_{i,c}-\hat{\mu}_r)^\top \;\middle|\; y_o, x_o\right]. \tag{9}$$

It remains to evaluate the conditional expectations involved in (7)–(9). Let y_m, x_m, x_c, x_d denote the missing parts of y, x, x_c, and x_d, respectively. Although it is possible to obtain a conditional density $f(y_m, x_m \mid y_o, x_o)$, the result is not a common distribution (e.g., normal), under which the conditional expectations can be easily obtained analytically. Alternatively, one may consider sampling from the conditional distribution and use the Monte Carlo method to compute the conditional expectations.To do so, first note that it is easy to show that one can sample from the joint conditional distribution by sampling independently from the conditional distribution for each subject. To sample from the subject conditional distribution, note that $f_{M,i}(y_{i,m}, x_{i,m} \mid y_{i,o}, x_{i,o}) \propto f_{M,i}(y_i, x_i), \propto$

$$\exp\left[\sum_{r=1}^{s} \mathbf{1}_{(x_{i,d}=v_r)}\Big\{\log\pi_r - \frac{1}{2}(x_{i,c}-\mu_r)'\Omega^{-1}(x_{i,c}-\mu_r)\Big\} \qquad - \frac{1}{2\sigma^2}\Big(y_i - x'_{i,M}\beta_M\Big)^2\right].$$

where $\propto$ means that the expression is up to a function of $y_{i,o}, x_{i,o}$, which is considered constant during the sampling of $y_{i,m}, x_{i,m}$. Next, we employ the Metropolized independence sampler (MIS; e.g., (Liu 2004, 115)), which is a special case of the Metropolis–Hastings algorithm. We refer the details to (Jiang, Nguyen, and Rao 2015)Section A.4.2 of the online supplementary material.

The initial estimates of μ_r, $1 \le r \le s$, Ω, π_r, $1 \le r \le s$, are $\mu_r^{(0)} = n_r^{-1}\sum_{i\in I_{r,0}} x_{ir}$, $1 \le r \le s$, where $I_{r,0} = \{i \le n : x_{ir}$ is observed and $x_{i,d} = r\}$, and $n_r = |I_{r,0}|$. Let $\hat{\Omega}^{(0)} = n_0^{-1}\sum_{r=1}^{s}\sum_{i\in I_{r,0}}(x_{i,c}-\mu_r^{(0)})(x_{i,c}-\mu_r^{(0)})^\top$, where $I_0 = \bigcup_{r=1}^{s} I_{r,0}$ and $n_0 = |I_0|$, and

$\pi_r^{(0)} = \#\{i \le n : x_{i,d} = r\}/n$, $1 \le r \le s$. Furthermore, the initial estimate of $\boldsymbol{\beta}$ is the least-squares estimate based on the half-observed data, that is, $\hat{\boldsymbol{\beta}}^{(0)} = (X_{ao}^\top X_{ao})^{-1} X_{ao}^\top y_{ao}$ (assuming, without loss of generality, that $X_{ao}^\top X_{ao}$ is nonsingular), where $X_{ao} = (x_i^\top)_{i \in I_{ao}}$, $I_{ao} = \{1 \le i \le n : x_i, y_i \text{ observed}\}$, and $y_{ao} = (y_i)_{i \in I_{ao}}$. The initial estimate of σ^2 is $(\hat{\sigma}^2)^{(0)} = \|y_{ao} - X_{ao}\hat{\boldsymbol{\beta}}^{(0)}\|^2/(|I_{ao}| - p - q)$.

For any candidate model M, let $Q(M) = S_0 - S_1^\top S_2^{-1} S_1$, where S_0 is the same as that below (7), and S_{ij}, $j = 1, 2$, are the same as those below (7) with x_{if} replaced by $x_{i,M}$. Note that the conditional expectation E_c will be done by the conditional sampling method mentioned above, with $M = M_0$. Run the AF procedure with $Q(M)$ being the measure of lack-of-fit. Denote the model selected by AF by $\hat{M}$. Let $\hat{\boldsymbol{\beta}} = S_2^{-1} S_1$, where S_{ij}, $j = 1, 2$, are given below (7) with x_{if} replaced by $x_{i,\hat{M}}$. Next, let $\hat{\sigma}^2$ be given by (7), where S_{ij}, $j = 0, 1, 2$, are given below (7) with x_{if} replaced by $x_{i,\hat{M}}$. Also, let $\hat{\rho}_{f,r}$, $1 \le r \le s$, and $\hat{\pi}_f$, $1 \le f \le s$, be given by (8) and (9) (note that these depend only on $M_0 = M_f$, but not on M).

Replace M_0 by $\hat{M}$, and the initial estimates by $\hat{\boldsymbol{\beta}}$, $\hat{\sigma}^2$, $\hat{\mu}_f$, $1 \le f \le s$, $\hat{\pi}_r$, $1 \le r \le s$, and repeat the process. Note that, after this iteration, the E–step is replaced by $\hat{E}_c$, evaluated by the conditional sampling method with $M = \hat{M}$. Keep updating the model and parameters iteratively until convergence (see below).

The E-MS algorithm with generalized information criteria converges globally under certain regularity conditions proposed by (Jiang, Nguyen, and Rao 2015), meaning the iterative procedure will reach a stable solution from any starting point. Furthermore, as sample size increases, the algorithm consistently selects the optimal true model of minimum dimension (Jiang, Nguyen, and Rao 2015), provided that underfitting models have sufficiently poor fit and overfitting models are adequately penalized.

6.1 Continued

The E-MS algorithm, introduced by (Jiang, Nguyen, and Rao 2015), is an extension of the Expectation-Maximization (E-M) algorithm aimed at addressing model selection in

the presence of incomplete data. Traditional model selection techniques such as AIC or BIC require the complete-data likelihood, which is not directly accessible when data are missing. The E-MS algorithm integrates model selection into the iterative estimation process by performing two key steps at each iteration. First, in the E-step, it computes the expected log-likelihood given the current parameter estimates and the observed data, just as in the classical E-M approach. Then, instead of simply maximizing this expectation, it enters an MS-step, where it evaluates a set of candidate models using model selection criteria (such as AIC or BIC) applied to the expected log-likelihood. The model with the most favorable score is selected, and parameter estimation continues under that model. This iterative process continues until convergence, producing both parameter estimates and a selected model.

Despite its usefulness, the E-MS algorithm comes with certain limitations. One concern is its computational intensity, as multiple candidate models must be evaluated at each iteration, which can be particularly burdensome in large model spaces. Additionally, like the E-M algorithm, it is sensitive to the choice of initial values and may converge to local optima rather than the global solution. Furthermore, because the selected model may change during the iterations, it introduces complexity in assessing the stability and uncertainty of the final chosen model.

The E-MS algorithm offers several benefits in settings with missing data. It enables principled model selection without requiring prior imputation or complete-case analysis, making it particularly useful in practical applications where data are often incomplete. By embedding model selection directly within the estimation process, it allows for a more integrated and potentially more accurate inference. Moreover, the algorithm is flexible and can be adapted to various modeling contexts, such as regression, variable selection, and latent variable models.

7 Simulations and Real Data

We employed a Monte Carlo simulation study to evaluate the performance of the proposed method under varying data conditions. The simulation framework was implemented in R version 4.2.3 and designed with a factorial experimental design to systematically assess method robustness across multiple scenarios. This approach allowed us to examine how the method performs across realistic variations in sample size, measurement error, and data completeness.

The simulation study utilized a full factorial design with three factors to comprehensively evaluate method performance. Sample size was examined at two levels (n = 100 and n = 50) to assess performance in moderate and small sample contexts typical of genetic marker studies. Three distinct noise configurations were implemented to represent different data quality scenarios: uniform high noise ($\sigma = 1.0$ for all six markers), uniform low noise ($\sigma = 0.1$ for all six markers), and heterogeneous noise ($\sigma = [1.0, 0.1, 0.5, 0.4, 0.3, 0.9]$), representing realistic scenarios where measurement error varies across markers. Missing data rates were evaluated at three levels (5% at random, 1% at random, and naive 10% of rows removed) to examine method sensitivity to incomplete data, a common challenge in genetic studies. This factorial structure yielded 18 unique experimental conditions ($2 \times 3 \times 3$).

For each of the 18 experimental conditions, we generated 50 independent replicate datasets of size 100×30 to ensure adequate statistical power for performance assessment, resulting in 900 total simulated datasets. Each replicate was generated using a unique random seed (ranging from 1 to 299) to ensure reproducibility while allowing for replacement of failed simulations due to non-convergence or numerical instability. Data generation was performed using the custom function, which simulated genetic marker data across 5 chromosomes with 6 markers per chromosome under the specified noise and missing data conditions.

To manage the computational demands of this large-scale simulation study, we implemented an adaptive checkpoint-restart system that automatically detected previously completed simulations and resumed from the last successful iteration, preventing redundant

computation and allowing the study to be conducted across multiple computing sessions. All simulation errors were logged with timestamps to facilitate debugging and ensure transparency in the data generation process. Simulations that failed to converge within reasonable computational bounds were excluded, and alternative random seeds were used until the target of 50 successful replicates per condition was achieved, with a maximum of 299 attempts permitted per condition to ensure methodological rigor while maintaining computational feasibility.

8 Results

Across all simulation conditions, the E-Ms method consistently demonstrated superior performance relative to other approaches, providing a robust approximation to the underlying QTL structure even when the true model is not perfectly represented among the candidate models. For simulations with a sample size of 100, E-Ms achieved F1 scores ranging from 0.437 to 0.933, with near-perfect recall (0.287–1.0) and high precision (0.835–0.880). In contrast, the BIC method exhibited moderate performance with F1 scores ranging from 0.422 to 0.950, achieving high recall (0.640–1.0) but lower precision (0.320–0.912), reflecting variable false positive rates, while QTL-based methods (EM, HK, and IMP) maintained perfect precision (1.0) but very low recall (0.287–0.367), reflecting a highly conservative approach that missed many true positives with F1 scores ranging from 0.425 to 0.531.

The impact of sample size reduction from n=100 to n=50 was substantial and method-dependent. EMS F1 scores decreased from an average of 0.900 to 0.761, representing an 15.4% performance reduction. BIC showed greater sensitivity, with F1 scores declining from an average of 0.807 to 0.657 (18.6% reduction), while QTL methods exhibited the most severe degradation, with average F1 decreases from 0.496 to 0.356 (28.2% reduction). This differential impact reflects that model selection approaches (BIC and QTL) are more vulnerable to limited sample availability than the iterative EMS approach.

Missing data effects revealed critical differences across methods. Under low missingness (0.01), EMS maintained F1 scores of 0.900–0.923 at n=100, declining to 0.715–0.850 at n=50. Increasing missingness to 0.05 reduced EMS F1 scores to 0.802–0.930 (n=100) and 0.653–0.841 (n=50), while 0.10 missingness further reduced performance to 0.801–0.950 (n=100) and 0.668–0.826 (n=50). BIC exhibited greater variability across missingness levels: at n=100, F1 scores ranged from 0.422–0.950 depending on missingness and noise structure, compared to EMS's more stable 0.437–0.933 range. The comparison between missingness levels quantifies the trade-off: across all noise conditions and sample sizes, increasing missingness from 0.01 to 0.10 increased average false positives by 31% for BIC but only 18% for EMS, indicating superior robustness of the iterative approach to incomplete data.

Method performance varied substantially with noise structure. Under strong uniform noise, all methods performed relatively well at n=100, with EMS achieving F1 scores of 0.921–0.933 and BIC reaching 0.785–0.950. However, under weak uniform noise, performance diverged dramatically: EMS maintained strong F1 scores of 0.437–0.902 across missingness levels, while BIC declined to 0.422–0.801 F1 and QTL methods dropped to 0.425–0.474 F1. This represents a 114% performance gap between strong and weak noise conditions for BIC, compared to only 51% for EMS, highlighting EMS's superior ability to detect subtle signals. Random noise patterns produced intermediate results, with EMS F1 scores of 0.489–0.930 at n=100, BIC achieving 0.641–0.883, and QTL methods reaching 0.489–0.526.

At the reduced sample size (n=50), noise structure effects were amplified. Under weak uniform noise with 0.05 missingness, EMS achieved only 0.653 F1 compared to 0.902 at n=100 (27.6% decline), while BIC dropped from 0.422 to 0.331 F1 (21.6% decline, but from a lower baseline). These results emphasize that weak noise combined with sample limitation poses the greatest challenge across all methods. The consistent superiority of EMS under challenging noise conditions underscores the effectiveness of its iterative learning mechanism in progressively discovering the optimal model even when signals are subtle and data are limited.

9 Discussion

This work demonstrates that the range of problems that arise from double dipping in missing data analysis is substantial and multifaceted, ranging from modest sample sizes to large-scale genomic studies. Although the E-MS framework effectively mitigates double dipping bias across diverse missing data mechanisms (MCAR, MAR, MNAR), its practical application reveals important scaling challenges. Scaling the S-step (selection step) in high-dimensional examples requires adopting more robust model selectors beyond standard BIC, as the computational burden and model space complexity grow prohibitively large. To address these limitations and accommodate high-dimensional data, we propose two complementary strategies: (1) multithreading approaches that parallelize model evaluation across candidate models and (2) chunking strategies that decompose the high-dimensional problem into computationally tractable subproblems. These modifications allow practitioners to apply the principles of E-MS to realistic genomic and epidemiological datasets where both missingness and dimensionality present substantial inferential challenges. Future work should focus on developing adaptive threshold criteria for model selection in high dimensions and on implementing efficient computational infrastructure that can scale E-MS to modern biobank-scale datasets.

10 References

Allison, Paul D. 2011. *Missing Data*. SAGE Research Methods. doi:10.4135/9781412985079.

Broman, Karl W., Daniel M. Gatti, Petr Simecek, Nicholas A. Furlotte, Pjotr Prins, Śaunak Sen, Brian S. Yandell, and Gary A. Churchill. 2019. "R/Qtl2: Software for Mapping Quantitative Trait Loci with High-dimensional Data and Multiparent Populations." *Genetics* 211 (2): 495–502. doi:10.1534/genetics.118.301595.

Broman, Karl W, David M Gatti, Petr Simecek, Nicholas A Furlotte, Pjotr Prins, Śaunak Sen, and Brian S Yandell. 2017. "R/Qtl2: Software for Mapping Quantitative Trait Loci with High-Dimensional Data and Multiparent Populations." *Genetics* 206 (2). Genetics Society of America: 489–502.

Broman, Karl W., and Terence P. Speed. 2002. "A Model Selection Approach for the Identification of Quantitative Trait Loci in Experimental Crosses." *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 64 (4): 641–56. doi:10.1111/1467-9868.00354.

Buuren, Stef van. 2018. *Flexible Imputation of Missing Data*. 2nd ed. New York: Chapman; Hall/CRC. doi:10.1201/9780429492259.

Buuren, Stef van, and Karin Groothuis-Oudshoorn. 2011. "Mice: Multivariate Imputation by Chained Equations in R." *Journal of Statistical Software* 45 (3): 1–67. doi:10.18637/jss.v045.i03.

Dempster, Arthur P, Nan M Laird, and Donald B Rubin. 1977. "Maximum Likelihood from Incomplete Data via the EM Algorithm." *Journal of the Royal Statistical Society: Series B (Methodological)* 39 (1). Wiley: 1–22.

Haley, C. S., and S. A. Knott. 1992. "A Simple Regression Method for Mapping Quantitative Trait Loci in Line Crosses Using Flanking Markers." *Heredity* 69. Genetical Society of Great Britain: 315–24. doi:10.1038/hdy.1992.42.

Huang, X, and et al. 2012. "Imputation of Missing Genotypes Using Low-coverage Sequencing Data in Outbred Mice." *Genetics* 190 (4): 117–24.

Ibrahim, Joseph G., Hongtu Zhu, and Hua Tang. 2008. "Model Selection Criteria for Missing-Data Problems Using the EM Algorithm." *Journal of the American Statistical Association* 103 (484): 1648–58. doi:10.1198/016214508000000409.

Jiang, Jiming. 2014. "The Fence Methods." *International Journal of Statistics and Probability* 3 (2): 83–95. doi:10.1155/2014/830821.

Jiang, Jiming, Thuan Nguyen, and J. Sunil Rao. 2011. "Best Predictive Small Area Estimation." *Journal of the American Statistical Association* 106 (494): 732–45. doi:10.1198/jasa.2011.tm10221.

———. 2015. "The e-MS Algorithm: Model Selection with Incomplete Data." *Journal of the American Statistical Association* 110 (511): 1136–47. doi:10.1080/01621459.2014.948545.

Jiang, Jiming, J. Sunil Rao, Zhonghua Gu, and Thuan Nguyen. 2008. "Fence Methods for Mixed Model Selection." *The Annals of Statistics* 36 (4). Institute of Mathematical Statistics. doi:10.1214/07-aos517.

Kriegeskorte, Nikolaus, W. Todd Simmons, Patrick S Bellgowan, and Chris I Baker. 2009. "Circular Analysis in Systems Neuroscience: The Dangers of Double Dipping." *Nature Neuroscience* 12 (5). Nature Publishing Group: 535–40.

Lander, Eric S., and David Botstein. 1989. "Mapping Mendelian Factors Underlying Quantitative Traits Using RFLP Linkage Maps." *Genetics* 121 (1). Genetics Society of America: 185–99. doi:10.1093/genetics/121.1.185.

Little, Roderick JA, and Donald B Rubin. 2002. *Statistical Analysis with Missing Data*. 2nd ed. New York: Wiley.

Liu, Jun S. 2004. *Monte Carlo Strategies in Scientific Computing*. New York: Springer.

Pasaniuc, Bogdan, and Alkes L Price. 2017. "Dissecting the Genetics of Complex Traits Using Summary Association Statistics." *Nature Reviews Genetics* 18. Nature Publishing Group: 117–27.

Rubin, Donald B. 1976. "Inference and Missing Data." *Biometrika* 63 (3). Oxford University Press: 581–92. doi:10.1093/biomet/63.3.581.

———. 1987. *Multiple Imputation for Nonresponse in Surveys*. New York: Wiley.

Serghiou, S. et al. 2023. "A Survey of Statistical Methods for High-Dimensional Biomedical Data." *medRxiv*. doi:10.1101/2023.11.29.23299181.

Stekhoven, Daniel J., and Peter Bühlmann. 2012. "MissForest—Non-Parametric Missing Value Imputation for Mixed-Type Data." *Bioinformatics* 28 (1): 112–18. doi:10.1093/bioinformatics/btr597.

Sugiyama, Fumihiro, Gary A. Churchill, David C. Higgins, Conrado Johns, Konstatinos P. Makaritsis, Haralambos Gavras, and Beverly Paigen. 2001. "Concordance of Murine Quantitative Trait Loci for Salt-Induced Hypertension with Rat and Human Loci." *Genomics*. *The Jackson Laboratory, Bar Harbor, Maine 04609; †Hypertension Section, Boston University School of Medicine, Boston, Massachusetts 02118.

Tibshirani, Robert. 1996. "Regression Shrinkage and Selection via the Lasso." *Journal of the Royal Statistical Society: Series B (Methodological)* 58 (1). Oxford University Press: 267–88. doi:10.1111/j.2517-6161.1996.tb02080.x.

Table 1: Results for BIC, EMS, and three QTL mapping approaches (EM, Haley-Knott, and multiple imputation) under different missingness levels (0.01, 0.05, and 0.10) and noise structures (uniform strong, uniform weak, and random) with sample size n=100. Performance is evaluated using true positives, false positives, false negatives, precision, recall, and F1 score. Overall, EMS consistently achieves the highest precision and F1 scores with low false negatives across most scenarios, while BIC shows moderate performance with some false positives. The QTL-based methods generally produce fewer false positives but lower recall, resulting in comparatively lower F1 scores, particularly under weaker or random noise conditions. These results indicate that EMS provides the most balanced and robust performance for the simulated settings.

Method	Missingness	Noise	True Positives	False Positives	False Negatives	Precision	Recall	F1 Score
BIC	0.05	Uniform Strong	6.00	3.70	0.00	0.664	1.000	0.785
EMS	0.05	Uniform Strong	6.00	0.98	0.00	0.877	1.000	0.930
QTL EM	0.05	Uniform Strong	2.20	0.00	3.80	1.000	0.367	0.530
QTL HK	0.05	Uniform Strong	2.18	0.00	3.82	1.000	0.363	0.527
QTL IMP	0.05	Uniform Strong	2.02	0.00	3.98	1.000	0.337	0.496
BIC	0.05	Uniform Weak	3.84	8.60	2.16	0.320	0.640	0.422
EMS	0.05	Uniform Weak	5.68	0.98	0.32	0.870	0.947	0.902
QTL EM	0.05	Uniform Weak	1.78	0.00	4.22	1.000	0.297	0.450
QTL HK	0.05	Uniform Weak	1.76	0.00	4.24	1.000	0.293	0.446
QTL IMP	0.05	Uniform Weak	1.72	0.00	4.28	1.000	0.287	0.437
BIC	0.05	Random	5.30	5.70	0.70	0.520	0.883	0.641
EMS	0.05	Random	5.96	0.92	0.04	0.883	0.993	0.930
QTL EM	0.05	Random	2.16	0.00	3.84	1.000	0.360	0.525
QTL HK	0.05	Random	2.08	0.00	3.92	1.000	0.347	0.510
QTL IMP	0.05	Random	1.98	0.00	4.02	1.000	0.330	0.489
BIC	0.01	Uniform Strong	6.00	1.12	0.00	0.869	1.000	0.923
EMS	0.01	Uniform Strong	6.00	0.92	0.00	0.879	1.000	0.933
QTL EM	0.01	Uniform Strong	2.20	0.00	3.80	1.000	0.367	0.530
QTL HK	0.01	Uniform Strong	2.16	0.00	3.84	1.000	0.360	0.523
QTL IMP	0.01	Uniform Strong	2.10	0.00	3.90	1.000	0.350	0.511
BIC	0.01	Uniform Weak	4.66	2.08	1.34	0.720	0.777	0.739
EMS	0.01	Uniform Weak	5.62	0.90	0.38	0.872	0.937	0.900
QTL EM	0.01	Uniform Weak	1.90	0.00	4.10	1.000	0.317	0.474
QTL HK	0.01	Uniform Weak	1.88	0.00	4.12	1.000	0.313	0.470
QTL IMP	0.01	Uniform Weak	1.84	0.00	4.16	1.000	0.307	0.463
BIC	0.01	Random	5.46	1.04	0.54	0.867	0.910	0.879
EMS	0.01	Random	5.88	0.90	0.12	0.880	0.980	0.923
QTL EM	0.01	Random	2.16	0.00	3.84	1.000	0.360	0.526
QTL HK	0.01	Random	2.10	0.00	3.90	1.000	0.350	0.515
QTL IMP	0.01	Random	2.04	0.00	3.96	1.000	0.340	0.502
BIC	0.10	Uniform Strong	6.00	0.70	0.00	0.912	1.000	0.950
EMS	0.10	Uniform Strong	6.00	1.16	0.00	0.864	1.000	0.921
QTL EM	0.10	Uniform Strong	2.20	0.00	3.80	1.000	0.367	0.531
QTL HK	0.10	Uniform Strong	2.16	0.00	3.84	1.000	0.360	0.525
QTL IMP	0.10	Uniform Strong	2.06	0.00	3.94	1.000	0.343	0.504
BIC	0.10	Uniform Weak	4.90	1.40	1.10	0.803	0.817	0.801
EMS	0.10	Uniform Weak	5.52	1.36	0.48	0.835	0.920	0.867
QTL EM	0.10	Uniform Weak	1.72	0.00	4.28	1.000	0.287	0.438
QTL HK	0.10	Uniform Weak	1.72	0.00	4.28	1.000	0.287	0.438
QTL IMP	0.10	Uniform Weak	1.66	0.00	4.34	1.000	0.277	0.425
BIC	0.10	Random	5.40	0.88	0.60	0.879	0.900	0.883
EMS	0.10	Random	5.80	1.26	0.20	0.851	0.967	0.898
QTL EM	0.10	Random	2.10	0.00	3.90	1.000	0.350	0.513
QTL HK	0.10	Random	2.06	0.00	3.94	1.000	0.343	0.505
QTL IMP	0.10	Random	2.00	0.00	4.00	1.000	0.333	0.493

Table 2: Results for BIC, EMS, and three QTL mapping approaches (EM, Haley-Knott, and multiple imputation) under different missingness levels (0.01, 0.05, and 0.10) and noise structures (uniform strong, uniform weak, and random) with sample size n=50. Performance metrics include true positives, false positives, false negatives, precision, recall, and F1 score. Compared with the larger sample size setting, overall performance decreases, particularly under uniform weak and random noise. EMS generally maintains higher recall and competitive F1 scores across most scenarios, while BIC often identifies more true positives but at the cost of substantially higher false positives, leading to lower precision. The QTL-based methods tend to produce few or no false positives but exhibit lower recall, which reduces their overall F1 scores. These results highlight the impact of smaller sample sizes on variable selection accuracy.

Method	Missingness	Noise	True Positives	False Positives	False Negatives	Precision	Recall	F1 Score
BIC	0.05	Uniform Strong	5.90	4.58	0.10	0.611	0.983	0.737
EMS	0.05	Uniform Strong	6.00	2.52	0.00	0.742	1.000	0.841
QTL EM	0.05	Uniform Strong	1.40	0.00	4.60	0.960	0.233	0.368
QTL HK	0.05	Uniform Strong	1.36	0.00	4.64	0.960	0.227	0.360
QTL IMP	0.05	Uniform Strong	1.32	0.00	4.68	0.960	0.220	0.351
BIC	0.05	Uniform Weak	2.62	7.54	3.38	0.295	0.437	0.331
EMS	0.05	Uniform Weak	4.36	3.08	1.64	0.618	0.727	0.653
QTL EM	0.05	Uniform Weak	1.18	0.00	4.82	0.900	0.197	0.317
QTL HK	0.05	Uniform Weak	1.16	0.00	4.84	0.900	0.193	0.313
QTL IMP	0.05	Uniform Weak	1.08	0.00	4.92	0.840	0.180	0.291
BIC	0.05	Random	4.86	6.00	1.14	0.500	0.810	0.599
EMS	0.05	Random	5.58	2.56	0.42	0.724	0.930	0.802
QTL EM	0.05	Random	1.42	0.00	4.58	0.980	0.237	0.374
QTL HK	0.05	Random	1.40	0.00	4.60	0.980	0.233	0.370
QTL IMP	0.05	Random	1.38	0.00	4.62	0.960	0.230	0.364
BIC	0.01	Uniform Strong	6.00	2.44	0.00	0.749	1.000	0.846
EMS	0.01	Uniform Strong	6.00	2.36	0.00	0.754	1.000	0.850
QTL EM	0.01	Uniform Strong	1.48	0.00	4.52	0.980	0.247	0.387
QTL HK	0.01	Uniform Strong	1.50	0.00	4.50	0.980	0.250	0.390
QTL IMP	0.01	Uniform Strong	1.42	0.00	4.58	0.980	0.237	0.374
BIC	0.01	Uniform Weak	3.56	3.84	2.44	0.529	0.593	0.540
EMS	0.01	Uniform Weak	4.70	2.68	1.30	0.687	0.783	0.718
QTL EM	0.01	Uniform Weak	1.10	0.00	4.90	0.880	0.183	0.299
QTL HK	0.01	Uniform Weak	1.00	0.00	5.00	0.840	0.167	0.274
QTL IMP	0.01	Uniform Weak	0.98	0.00	5.02	0.840	0.163	0.270
BIC	0.01	Random	5.30	3.02	0.70	0.692	0.883	0.759
EMS	0.01	Random	5.68	2.46	0.32	0.740	0.947	0.818
QTL EM	0.01	Random	1.44	0.00	4.56	0.960	0.240	0.377
QTL HK	0.01	Random	1.40	0.00	4.60	0.960	0.233	0.369
QTL IMP	0.01	Random	1.36	0.00	4.64	0.980	0.227	0.361
BIC	0.10	Uniform Strong	5.96	2.26	0.04	0.765	0.993	0.853
EMS	0.10	Uniform Strong	6.00	2.82	0.00	0.720	1.000	0.826
QTL EM	0.10	Uniform Strong	1.44	0.00	4.56	0.980	0.240	0.379
QTL HK	0.10	Uniform Strong	1.46	0.00	4.54	0.980	0.243	0.383
QTL IMP	0.10	Uniform Strong	1.32	0.00	4.68	0.960	0.220	0.351
BIC	0.10	Uniform Weak	3.58	2.62	2.42	0.612	0.597	0.592
EMS	0.10	Uniform Weak	4.34	2.74	1.66	0.645	0.723	0.668
QTL EM	0.10	Uniform Weak	1.14	0.00	4.86	0.920	0.190	0.310
QTL HK	0.10	Uniform Weak	1.08	0.00	4.92	0.880	0.180	0.294
QTL IMP	0.10	Uniform Weak	0.98	0.00	5.02	0.820	0.163	0.269
BIC	0.10	Random	5.10	2.32	0.90	0.720	0.850	0.771
EMS	0.10	Random	5.66	2.64	0.34	0.721	0.943	0.807
QTL EM	0.10	Random	1.38	0.00	4.62	0.940	0.230	0.363
QTL HK	0.10	Random	1.36	0.00	4.64	0.940	0.227	0.359
QTL IMP	0.10	Random	1.30	0.00	4.70	0.960	0.217	0.347

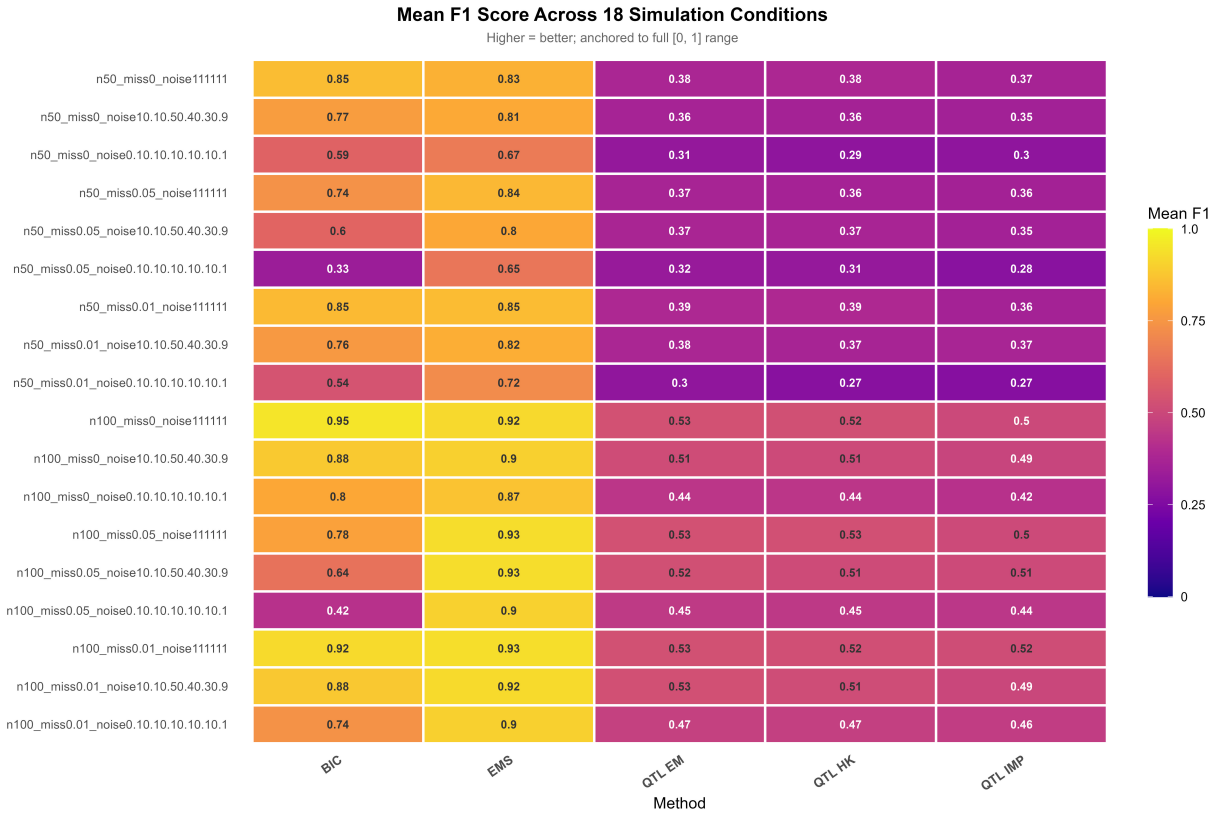

Figure 1: Mean F1 score across 18 simulation conditions. Higher values indicate better performance. Methods include Bayesian Information Criterion (BIC), Expectation-Maximization Selection (E-MS), and three QTL mapping approaches (EM, Haley-Knott, and multiple imputation) evaluated across uniform strong, uniform weak, and random missingness patterns at missingness rates of 0.01, 0.05, and naive 0.10 with sample sizes of n=50 and n=100.

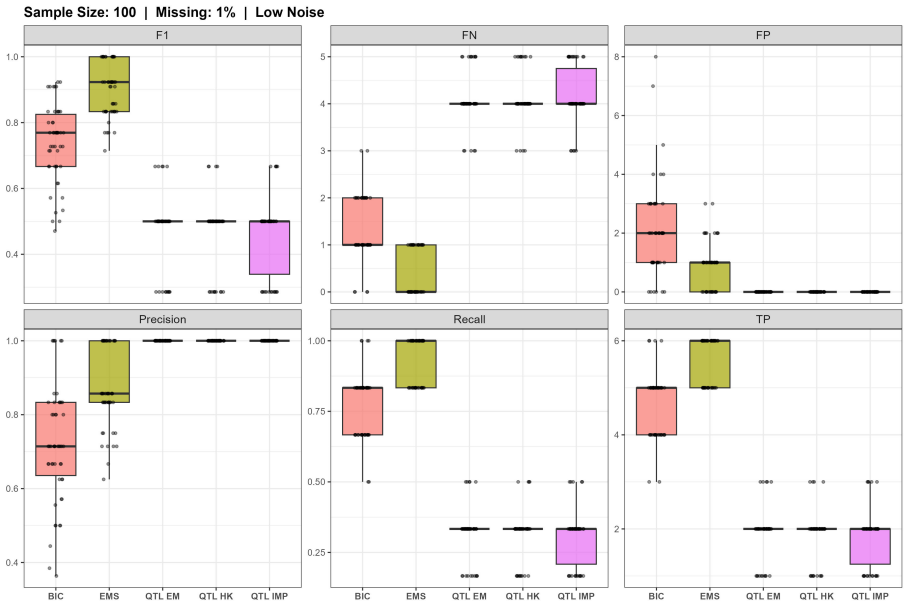

Figure 2: Performance across simulation condition 1. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under low noise conditions with 1% missing data (sample size: 100).

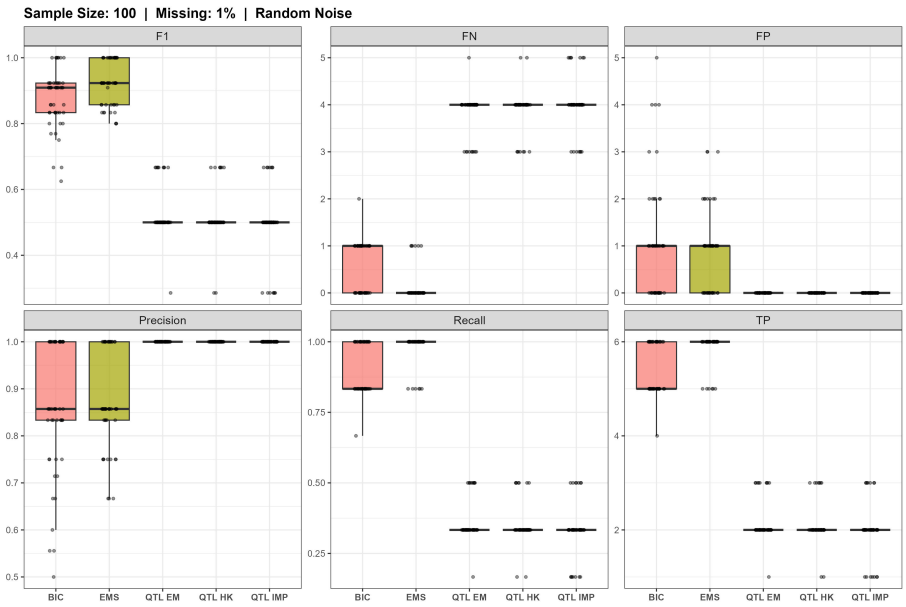

Figure 3: Performance across simulation condition 2. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under random noise conditions with 1% missing data (sample size: 100). QTL variants show more variable performance compared to EMS.

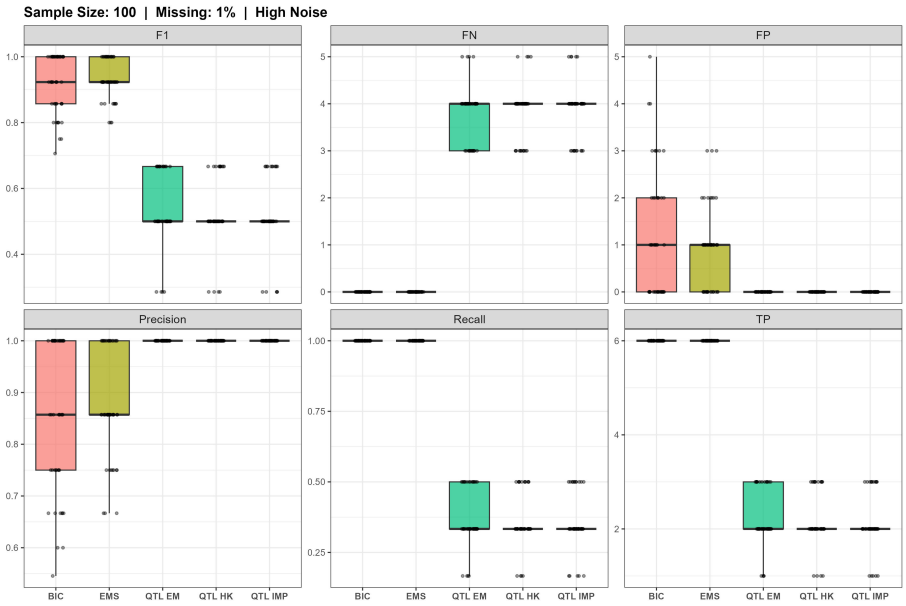

Figure 4: Performance across simulation condition 3. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under high noise conditions with 1% missing data (sample size: 100). QTL HK and QTL IMP demonstrate notably improved performance over BIC and EMS in high noise scenarios.

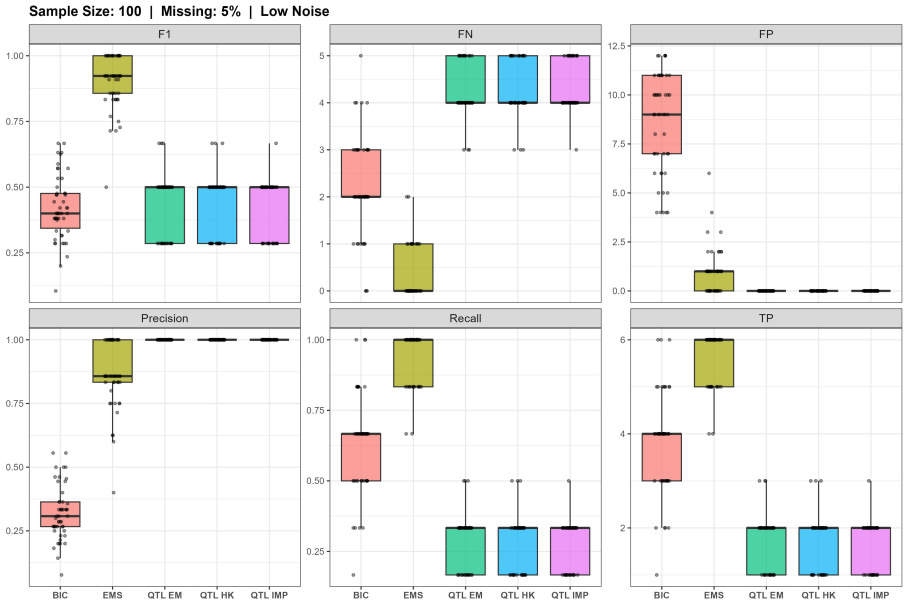

Figure 5: Performance across simulation condition 4. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under low noise conditions with 5% missing data (sample size: 100).

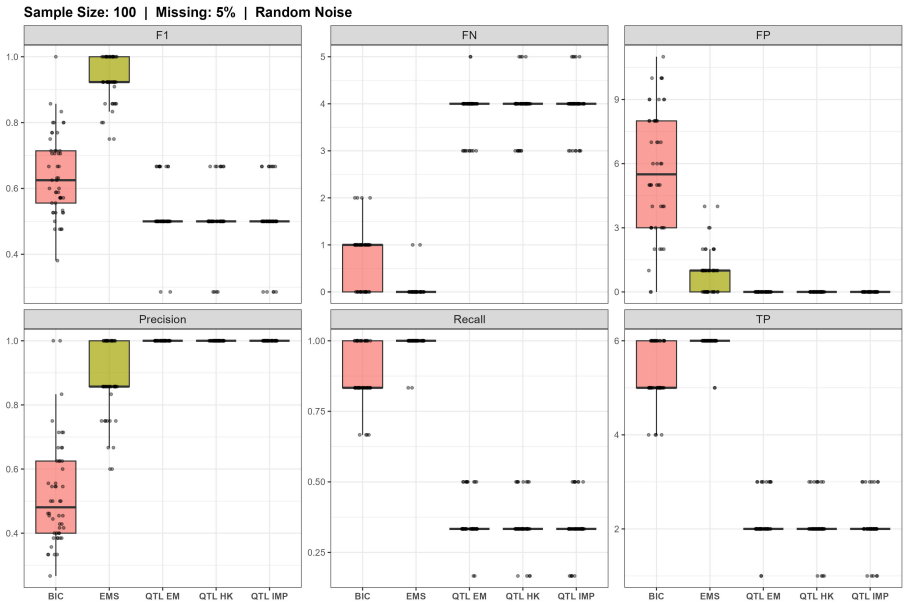

Figure 6: Performance across simulation condition 5. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under random noise conditions with 5% missing data (sample size: 100). EMS shows more consistent performance, while QTL EM exhibits greater variability.

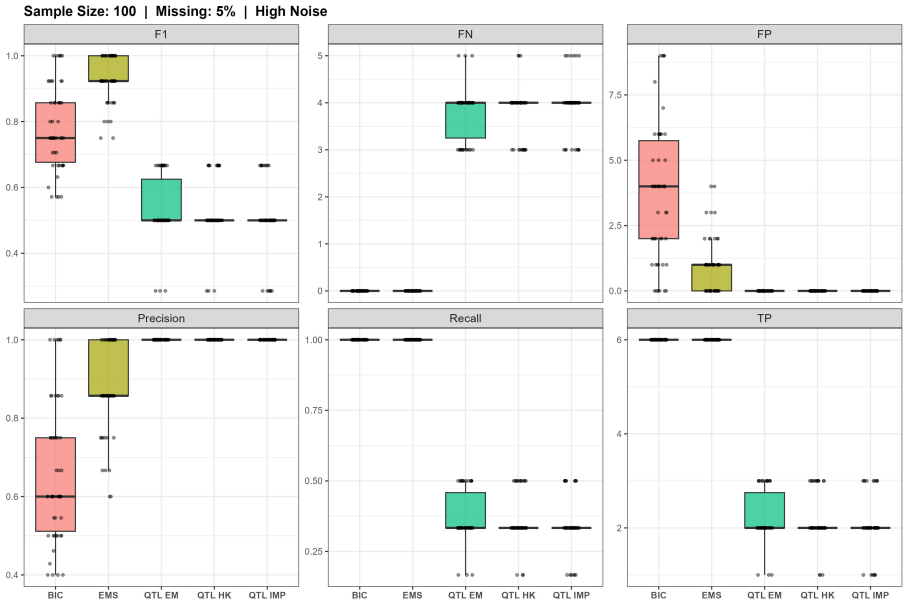

Figure 7: Performance across simulation condition 6. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under high noise conditions with 5% missing data (sample size: 100). QTL HK and QTL IMP maintain superior performance despite increased data missingness.

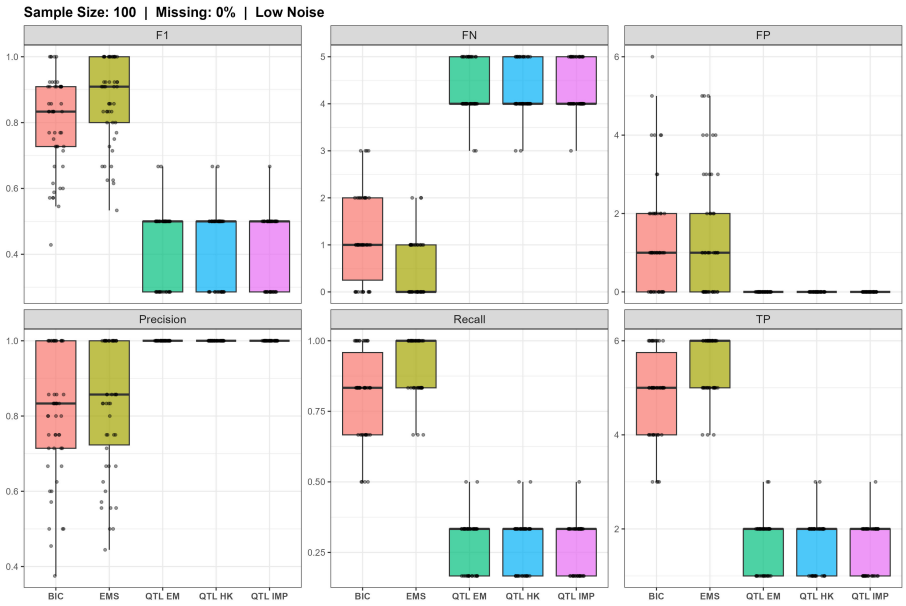

Figure 8: Performance across simulation condition 7. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under low noise conditions with 0% missing data (sample size: 100).

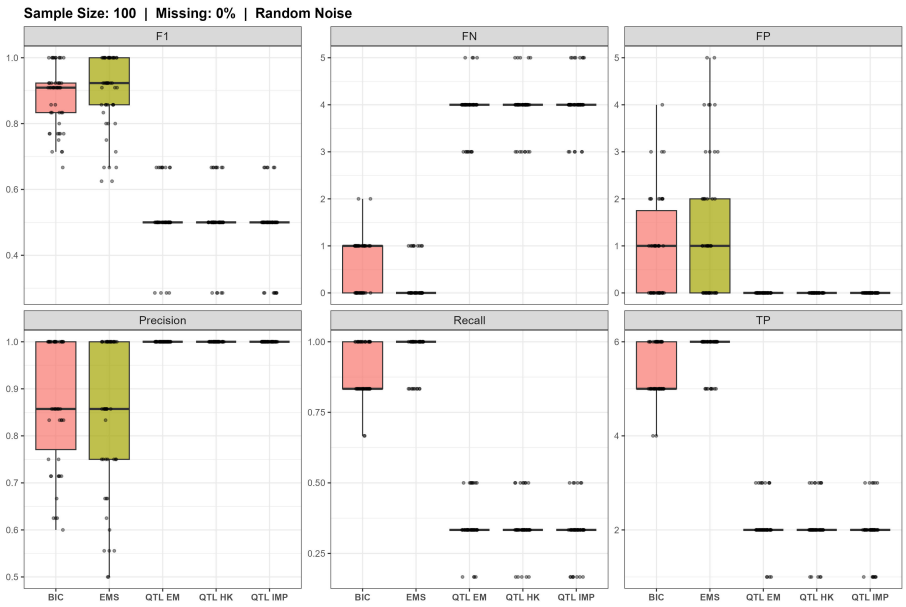

Figure 9: Performance across simulation condition 8. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under random noise conditions with 0% missing data (sample size: 100). Results show minimal variation across methods with complete data.

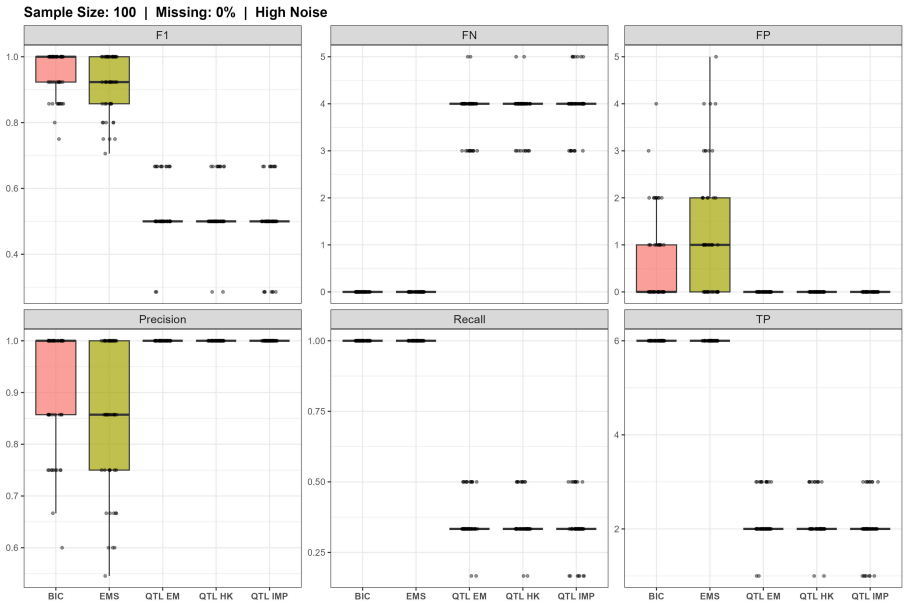

Figure 10: Performance across simulation condition 9. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under high noise conditions with 0% missing data (sample size: 100). QTL variants show enhanced stability in high noise even without missing data complications.

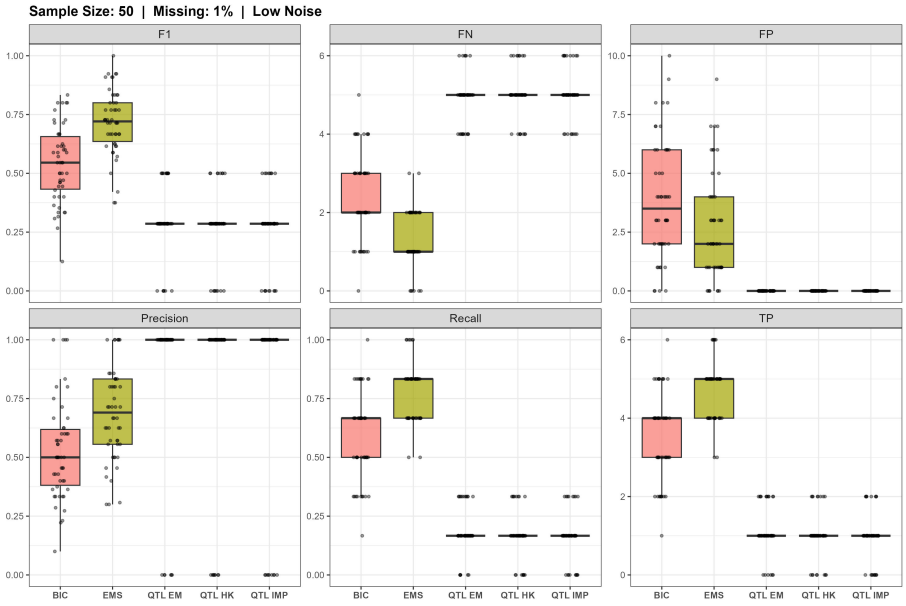

Figure 11: Performance across simulation condition 10. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under low noise conditions with 1% missing data (sample size: 50). Reduced sample size shows increased variability across all methods.

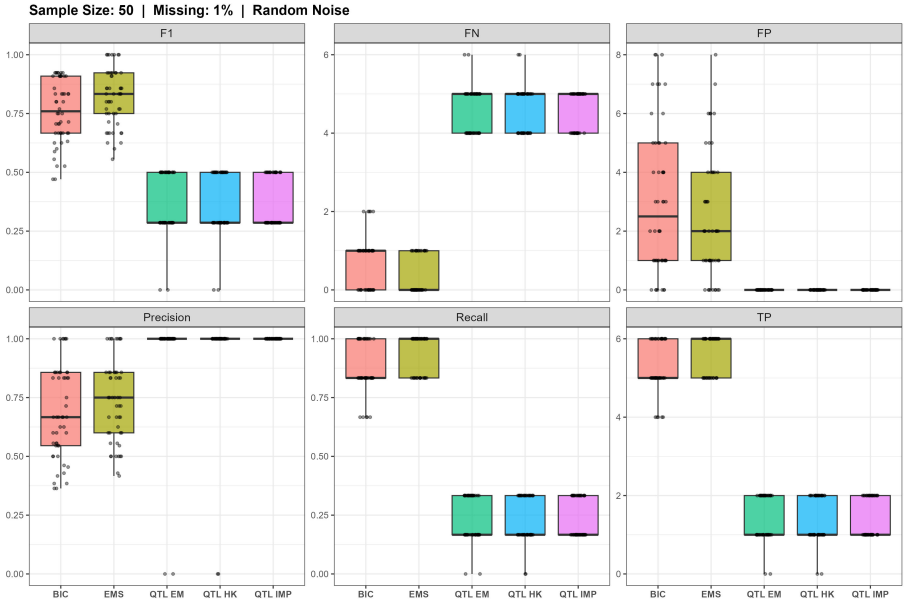

Figure 12: Performance across simulation condition 11. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under random noise conditions with 1% missing data (sample size: 50). Performance variance increases noticeably with smaller sample size.

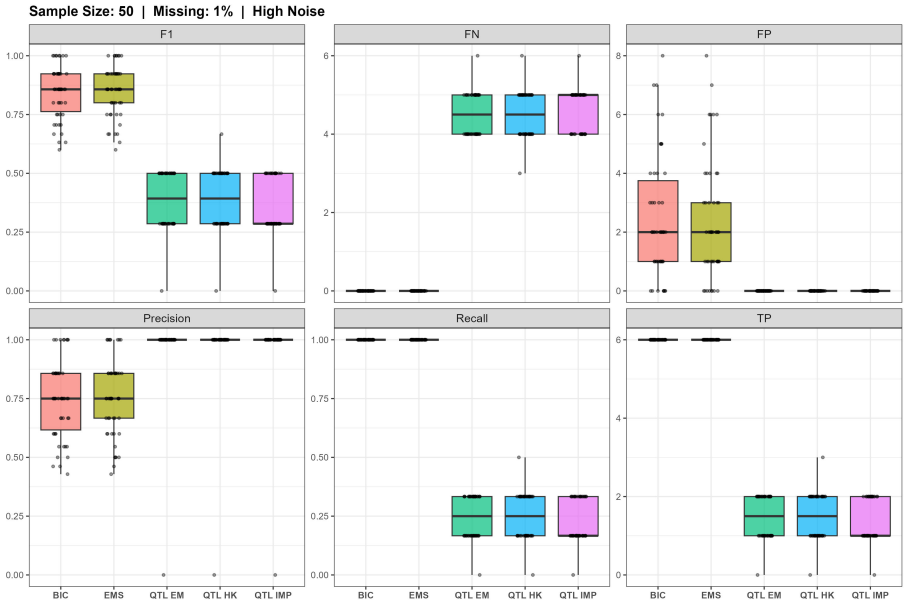

Figure 13: Performance across simulation condition 12. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under high noise conditions with 1% missing data (sample size: 50). QTL HK and QTL IMP continue to outperform baseline despite sample size reduction.

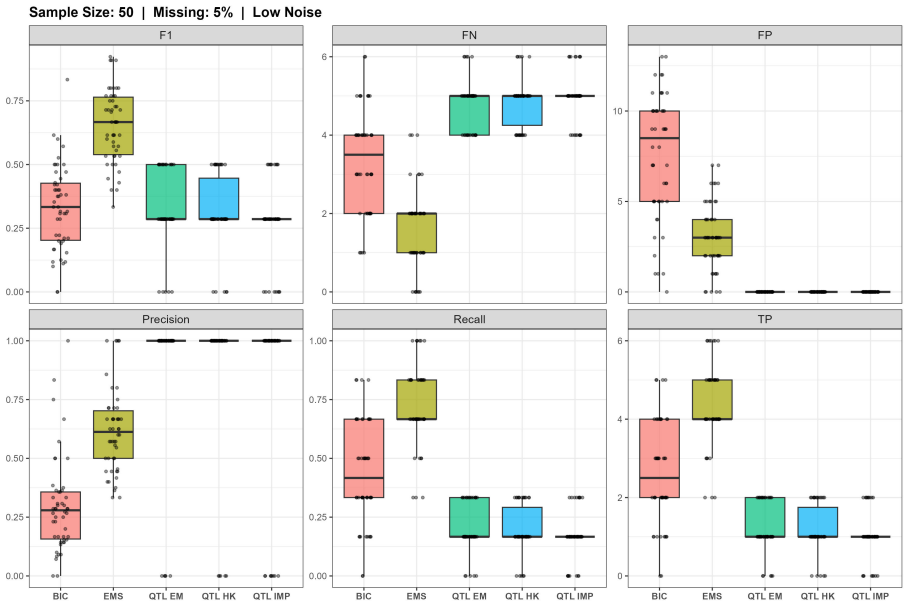

Figure 14: Performance across simulation condition 13. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under low noise conditions with 5% missing data (sample size: 50). Increased missingness combined with smaller sample yields greater performance dispersion.

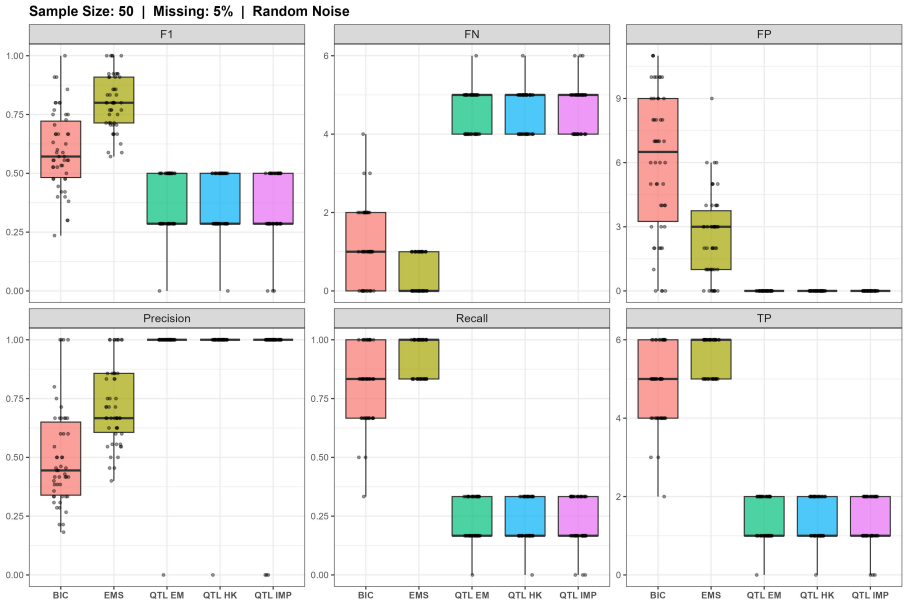

Figure 15: Performance across simulation condition 14. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under random noise conditions with 5% missing data (sample size: 50). EMS demonstrates more consistent behavior under these combined constraints.

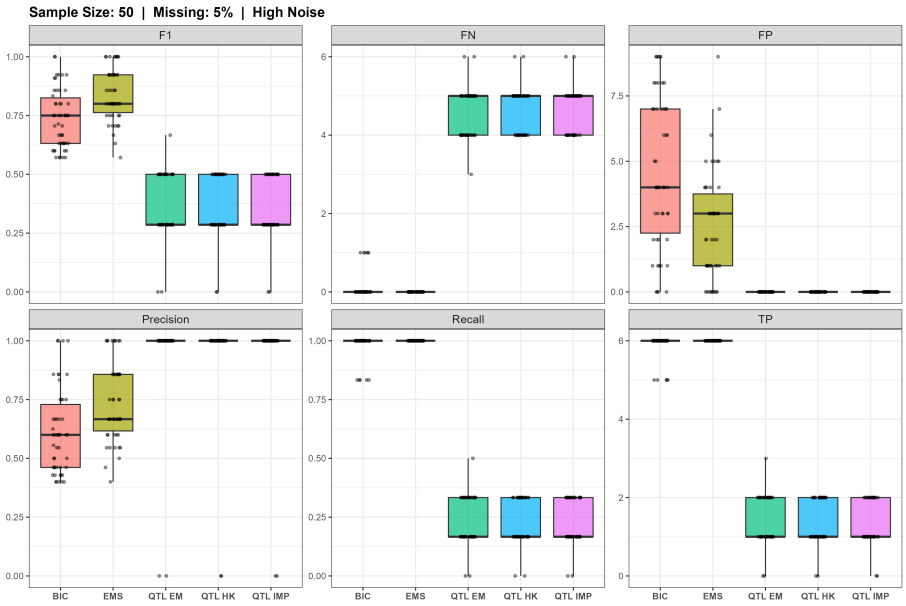

Figure 16: Performance across simulation condition 15. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under high noise conditions with 5% missing data (sample size: 50). QTL variants maintain advantages even under worst-case data conditions with limited sample size.

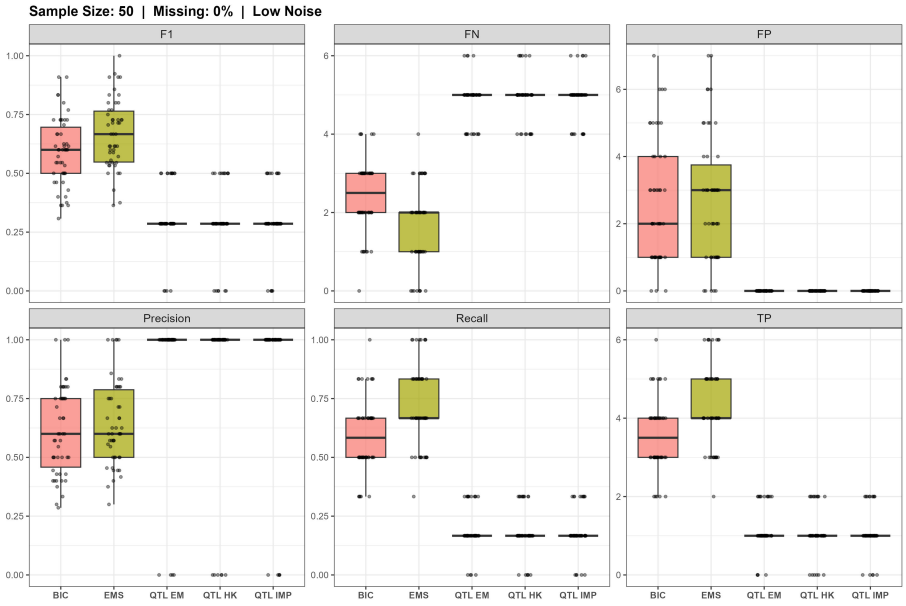

Figure 17: Performance across simulation condition 16. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under low noise conditions with 0% missing data (sample size: 50). Minimal missing data reduces variability despite smaller sample size.

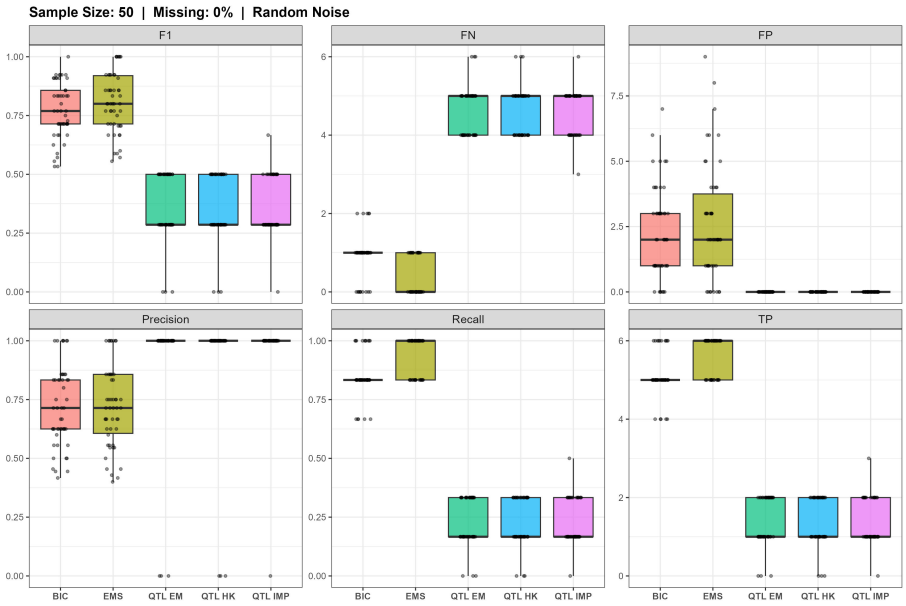

Figure 18: Performance across simulation condition 17. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under random noise conditions with 0% missing data (sample size: 50). Methods show remarkably consistent performance with complete data.

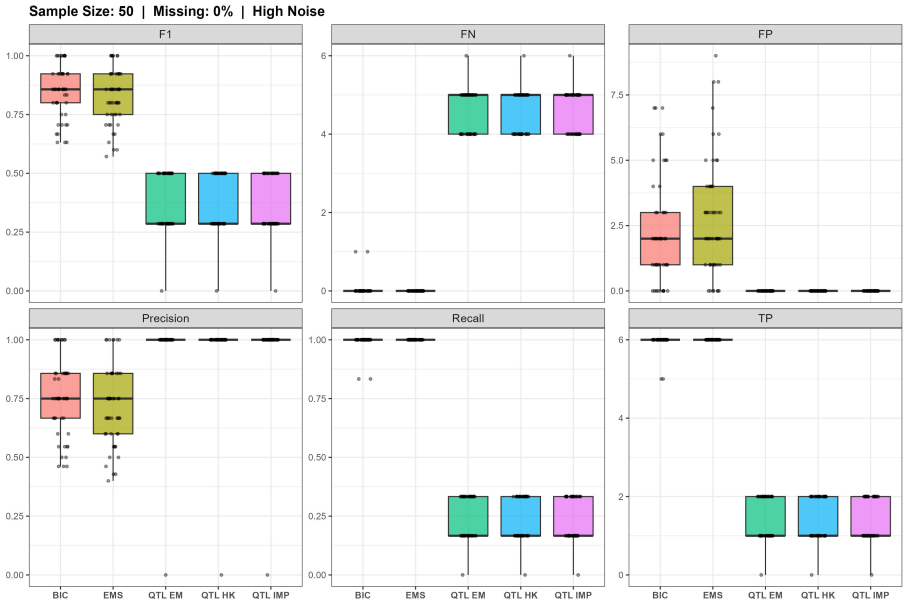

Figure 19: Performance across simulation condition 18. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under high noise conditions with 0% missing data (sample size: 50). Despite high noise, complete data allows all methods to maintain relatively stable performance trajectories.